Jonah Rosenblum

Education	Computer Science and Engineering University of Michigan	**Email**	jonaher@umich.edu
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Location	Ann Arbor, MI		

Research Interests

My recent work explores how secure systems can enforce data-use policies without relying solely on traditional trust anchors such as identity or credentials. I develop systems that use contextual signals, including time, location, and static or dynamic code invariants, to protect sensitive data in settings where users or AI agents are not fully trusted. I have also worked on data privacy in genomics, Rowhammer defenses, and scalable memory freshness.

Education

2022 – Now Ph.D. in Computer Science and Engineering
University of Michigan, Ann Arbor, MI
Advisor: Satish Narayanasamy
GPA: 3.99

2021 M.S. in Computer Science and Engineering
University of Michigan, Ann Arbor, MI

2017 – 2020 B.S. in Computer Science
University of Michigan, Ann Arbor, MI
GPA: 3.83

Employment History

May 2025 – Now Hewlett Packard Enterprise Labs, CA
Research Associate – Systems Architecture Lab

- Initiated new formal verification project to detect misconfiguration in security intent.
- Satisfied lab quarterly OKR with redesign of DAOS security architecture to work with the new authentication backend.

May – August 2021 Google, CA
Software Engineering Intern – Cloud Trace Team

Worked on open-source telemetry tool OpenTelemetry, implementing graceful shutdown for processes to ensure all traces and metrics are exported.

May – August 2020 Google, CA
Software Engineering Intern – GCloud Infrastructure Team

Analyzed inversion between network and application priority for high-priority Google traffic across all clusters and identified strategies to align less latency-sensitive traffic with appropriate QoS.

Publications

- **Jonah Rosenblum**, Juechu Dong, Peter Chen, Satish Narayanasamy. Timelock Drive: Isolated Time-Based Defense for Storage Systems. In USENIX Symposium on Operating Systems Design and Implementation (**OSDI**) 2026.
- **Jonah Rosenblum**, Juechu Dong, Satish Narayanasamy. Confidential computing for population-scale genome-wide association studies with SECRET-GWAS. In **Nature Computational Science**. 2025.
- Juechu Dong, **Jonah Rosenblum**, Satish Narayanasamy. Toleo: Scaling Freshness to Terascale Memory using CXL and PIM. In Architectural Support for Programming Languages and Operating Systems (**ASPLOS**). 2024.
- Kevin Loughlin, **Jonah Rosenblum**, Stefan Saroiu, Alec Wolman, Dimitrios Skarlatos, and Baris Kasikci. Siloz: Leveraging DRAM Isolation Domains to Prevent Inter-VM Rowhammer. In Symposium on Operating Systems Principles (**SOSP**). 2023.

Teaching

2024 Parallel Computer Architecture (EECS 570)
University of Michigan, Ann Arbor, MI

Jan – May 2024; Graduate Student Instructor for Prof. Ronald Dreslinski
Teaching evaluation: 4.8/5

2023 Advanced Operating Systems (EECS 582)
University of Michigan, Ann Arbor, MI

Sep – Dec 2023; Graduate Student Instructor for Prof. Ryan Huang
Teaching evaluation: 4.8/5
One of 3 out of 400+ student instructors to receive the **Outstanding GSI Award**

2021 Parallel Computer Architecture (EECS 570)
University of Michigan, Ann Arbor, MI

Jan – May 2021; Graduate Student Instructor for Prof. Satish Narayanasamy
Teaching evaluation: 4.8/5

Professional Activities

Service

2025 USENIX ATC 2025 Artifact Evaluation Committee

At Michigan

2025 – 2026 Faculty Search Committee Member
Ann Arbor, MI

Elected by all PhD and Master's students in the department to be the student representative serving on the faculty search committee for the 25/26 cycle.

2023 – Now CSEG Security Reading Group Co-Chair
Ann Arbor, MI

Since Jan 2023
Run weekly security group meetings to discuss current research papers.

2023 – Now Grad Mentor Program Volunteer
Ann Arbor, MI

Since Sep 2023
Research and graduate program mentor to Master's and Ph.D. students.

Technical Skills

- **Programming**

  Imperative programming (C, C++, Python, Golang, Rust, etc.) - Proficient
  Functional programming - Competent

- **Systems & Development**

  Kernel development - QEMU/Linux
  Formal verification - Dafny/Hoare logic